

Final Course Design Report

Community-Driven Housing Information Platform Addressing China's Property Market Challenge

A Web Application Response to the Property Crisis in China



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Course: J2EE-Based Framework for Enterprise

软件工程
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Chapter 1 — Project Topic Selection and Explanation

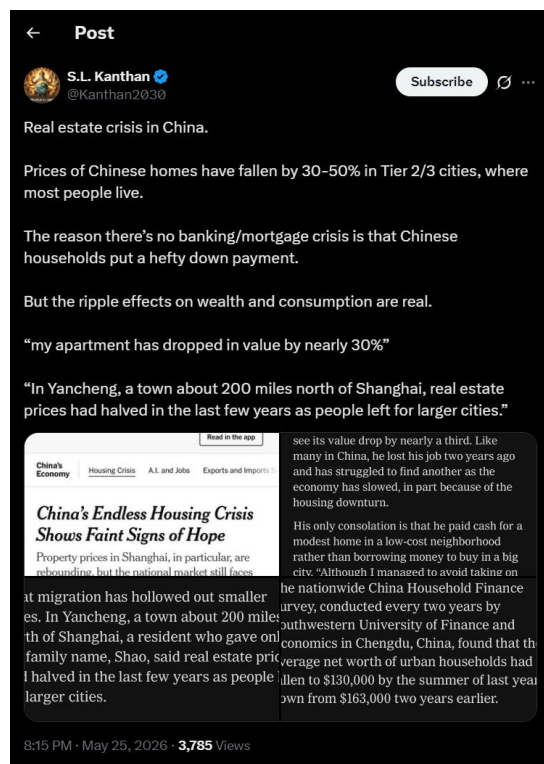
1.1 What Is This Project?

This project is a community-based housing discovery and review platform built for foreigners and young people living in Chinese cities. Users can browse affordable housing listings, read honest reviews from other tenants, and rate neighborhoods based on their real living experience. The platform is built using Java J2EE technologies with a MySQL database backend.

1.2 What Problem Does It Address?

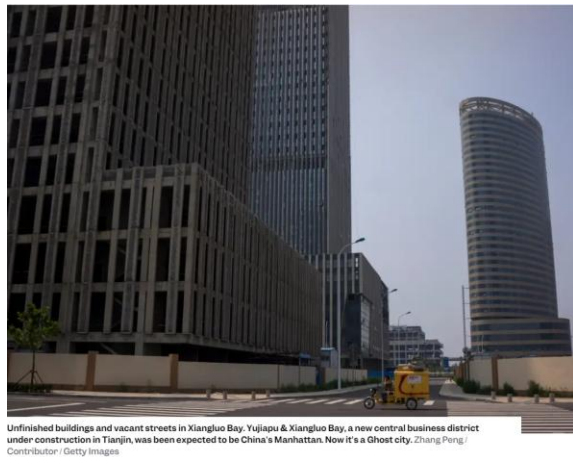
China is currently going through one of the worst property crises in its history. For roughly two decades, real estate was treated as the safest and most profitable investment in the country. Developers borrowed enormous amounts of money to build as fast as possible, and ordinary families poured their savings into apartments — not necessarily to live in, but because they believed property prices would always go up.

That belief has now broken. Major developers like Evergrande have collapsed under hundreds of billions of dollars in debt. Across the country, there are an estimated 65 to 80 million empty apartments that nobody is living in. Entire districts were built faster than the actual population needed them, and in many cities, new residential areas stand almost completely empty — what people now call "ghost cities."



(News coverage of China's property crisis. Source: X, 微博)

The crisis is made worse by two deeper issues. First, China's birth rate has been falling for years, which means there are fewer young families forming and therefore less natural demand for new housing. Second, many people who could afford to buy are simply choosing not to, because they are afraid prices will keep dropping. When enough people think that way, it becomes a self-fulfilling cycle: less buying, less confidence, lower prices, even less buying.



Unfinished buildings and vacant streets in Xiangluo Bay, Yujiapu & Xiangluo Bay, a new central business district under construction in Tianjin, was been expected to be China's Manhattan. Now it's a Ghost city. Zhang Peng / Contributor / Getty Images



(Empty residential buildings in China's ghost cities. Source: Google, 微博)

The result is a strange and painful situation. There are millions of empty apartments across China. At the same time, many people — especially young workers, migrants, and international students, are struggling to find affordable and trustworthy housing in the cities where they actually want to live. The supply exists. But the connection between available housing and the people who need it is broken.

A big part of why that connection is broken is trust. Most existing Chinese housing platforms are flooded with developer advertisements and inflated listings. There is very little honest, community-sourced information about what it is actually like to live in a specific apartment or neighborhood — especially from the perspective of foreigners, who face additional challenges like language barriers and unfamiliarity with local systems. Many international students end up finding housing through informal WeChat groups or by relying on whoever happens to help them when they arrive, rather than through any organized or reliable source of information.

This platform is built to address that specific gap. It is not trying to fix the entire property crisis — that would be unrealistic. But it aims to make one part of the problem smaller: helping people who genuinely need housing find, evaluate, and trust their options through honest community reviews and neighborhood ratings.

☆ 热门



董仲鑫-小董老师

2025年08月26日 11:16 来自 微博视频号

#恒大被取消上市地位# Evergrande, formerly one of China's biggest property developers, has been delisted from the Hong Kong stock exchange, marking another step in the retreat of the giant real estate developer whose downfall contributed to a prolonged crisis in China's property market. Evergrande's creditors are still working to wind up debts that amounted to more than \$340 billion. Once China's second-largest developer, it ran into trouble when Chinese regulators cracked down several years ago on what they deemed to be excess borrowing by developers. 董仲鑫-小董老师的微博视频 收起^



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2

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(The collapse of major Chinese property developers such as Evergrande. Source : Google 微博)

1.3 Why Is This Topic Worth Building?

It solves a real problem.

The housing information gap described above is happening right now. There is a clear group of people — foreign students, young migrants, first-time renters — who do not have a good way to find trusted housing information in China. This platform directly serves that group.

It is realistic to build.

The system does not require external APIs or complex integrations. All of its core features — user registration, login, housing listings, reviews, ratings, and neighborhood pages — can be built entirely with the technologies covered in this course, including Servlet, JSP, EL expressions, JSTL, JDBC, Filter, Session and Cookie management, and the five-tier architecture pattern with a MySQL database.

It has genuine social value.

By helping people navigate the housing market more confidently, the platform contributes — even in a small way — toward getting some of China's vacant housing stock into actual use. Every person who finds a good apartment through honest community information is one fewer empty unit sitting idle.

Evergrande's Fall : The largest bankruptcy in China's history

Evergrande's historic bankruptcy exposes China's property crisis, regulatory risks, global deflation threats, and the fragility of a debt-fuelled economic model.



December 12, 2025 Articles

By Dimitri Gellé

On 29 January 2024, a Hong Kong court handed down the verdict: the liquidation order of China Evergrande Group, a property giant once synonymous with China's economic boom. The developer's fall was not just a standalone bankruptcy, but a watershed moment that highlighted the viability of the Chinese economic model and its complex links to the global economy. On 25 August 2025, the company was to be officially delisted from the Hong Kong Stock Exchange [7]. The fall of Evergrande, burdened by a colossal debt of 2.39 trillion yuan (\$330 billion) – equivalent to 2% of the country's GDP [1], serves as a litmus test for Beijing's governance and shows that a national crisis can have international repercussions.

The Domestic Roots of a Global Crisis

Evergrande's extraordinary rise and fall is a perfect example of an economic philosophy that prioritised "unmitigated economic growth" [3]. For decades, the company thrived on a high-risk strategy: borrowing heavily to acquire land, then pre-selling apartment units to finance the real estate project [2]. This model transformed the property sector into a national economic pillar, accounting for up to 30% of China's Gross Domestic Product (GDP) and holding the vast majority of Chinese household wealth [1, 4].

To grasp the scale of this dependence, it is essential to note that Chinese households hold approximately 70% of their wealth in real estate [1, 4]. By comparison, the average for households in the euro area was around 50% in 1999, rising to approximately 62% in 2008 [10]. This disparity underscores the unique vulnerability of Chinese households to fluctuations in the property market.

It is personally relevant.

This project was inspired by my own personal experience. As an international student living in Nanjing, I have seen firsthand how difficult it is for foreigners to find reliable housing information without local connections or language support. When I visited Shenzhen, I was struck by how many fully built apartment buildings and commercial areas were standing almost completely empty — a visible reminder of just how real this crisis is. That experience is what gave this project its direction and purpose.

1.4 Research Orientation

This project sits at the intersection of two things: understanding a real social and economic problem, and applying Java development skills to build something that responds to it. The goal is to think through how software can be designed to serve people who are often left out of existing platforms — in this case, foreigners trying to find their footing in a Chinese city during a difficult period in the country's housing market.

The project also serves as a full demonstration of the J2EE technology stack studied throughout this course, from basic HTTP and Servlet fundamentals through to layered backend architecture and modern Spring Boot controller design. The aim is to build something that works well, makes sense as a product, and reflects the kind of thinking that goes into real-world web development.

Chapter 2 — Research and Analysis of Similar Projects

2.1 Overview

To better understand what already exists in the housing platform space and where the gaps are, three well-known platforms were researched and analyzed: Anjuke (安居客), Lianjia (链家), and Trulia. These three were chosen because they are the most relevant comparisons to this project — two from China and one from the West — and together they show both what current solutions offer and what they are still missing.

2.2 Anjuke (安居客)

What It Is

Anjuke is one of China's biggest online real estate platforms, founded in 2007 and now owned by 58.com. It covers new homes, second-hand homes, rental properties, and commercial real estate all in one place. It is primarily used by Chinese buyers, renters, and real estate agents.



(Anjuke (安居客) website homepage and Wechat mini program)

Architectural Model

Anjuke most likely follows a standard MVC (Model-View-Controller) architecture at its core, with a clear separation between the data layer, business logic, and front-end presentation. Given its scale, it likely uses a microservices-based backend where different services handle listings, search, user management, and payments independently. The front end is mobile-first, built as a native app with a web platform running in parallel.

Core Features

- Millions of property listings searchable by city, price range, property type, and size
- High-resolution photos, floor plans, and detailed descriptions for each listing
- Price trend data and market analysis tools
- VR home viewing and 3D tours for new developments
- In-app messaging between users and landlords or agents
- Mobile-first design with a widely used Android and iOS app

What It Does Well

Anjuke is very strong at aggregating a huge volume of listings in one place and making them easy to filter and browse. The search experience is smooth, and the amount of data available — price history, area comparisons, neighborhood maps — is genuinely useful for someone who already knows what they are looking for.

What It Is Missing

The biggest weakness of Anjuke is that it is entirely driven by advertisers and real estate agents. Almost all the content on the platform comes from developers and agents trying to sell or rent properties, which means the information is naturally biased. There are no genuine community reviews from tenants who have actually lived in these apartments. The platform is also entirely in Chinese, with no English or multilingual support — making it essentially unusable for international students and foreign workers without strong Mandarin skills.

2.3 Lianjia (链家)

What It Is

Lianjia was founded in 2001 in Beijing and has grown into China's leading real estate brokerage brand. Unlike Anjuke, which is primarily a listings platform, Lianjia combines online listings with an offline network of physical stores and trained agents.



(Lianjia (链家) website homepage)

Architectural Model

Lianjia most likely uses an Online-to-Offline (O2O) architecture, meaning its digital platform is tightly integrated with its physical branch network. The backend follows a layered architecture similar to the five-tier model — with a presentation layer, business logic layer, data access layer, and database — extended with real-time agent coordination tools. Its parent platform Beike (贝壳) acts as the broader ecosystem layer on top.

Core Features

- Verified property listings confirmed through agent inspection and ownership checks
- A large offline network of physical branches across major Chinese cities
- An Agent Cooperation Network (ACN) that allows agents to share commissions
- Transaction support tools including deposit guarantees and fund safekeeping
- Price trend data and market insights

What It Does Well

Lianjia's main strength is trust and verification. Unlike most listing platforms where anyone can post anything, Lianjia has agents who physically inspect properties and verify ownership before listings go live. Their transaction support features — like protecting deposits and ensuring money is held safely during deals — also give users more confidence when making a major purchase.

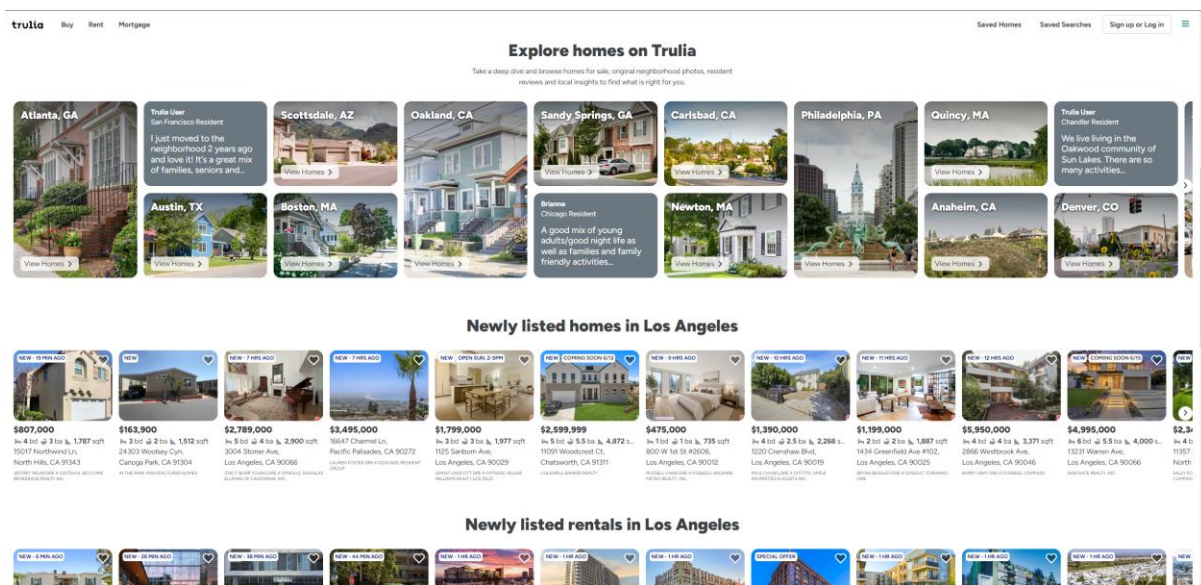
What It Is Missing

Lianjia is built around the buying and selling process, with professional agents at the center of every interaction. This makes it less useful for ordinary renters who just want to find a decent apartment. Like Anjuke, there is no space for community-sourced information. Tenants cannot leave reviews about landlord behavior, an apartment's actual condition, or what daily life in a neighborhood feels like. It is also entirely built for Chinese users, with no English interface or foreigner-specific guidance.

2.4 Trulia (United States)

What It Is

Trulia is the largest real estate platform in the United States, launched in 2006. While it operates in a completely different market from China, it is the most relevant international reference for this project because it introduced community-based features into the housing search experience — something that does not yet exist in China's housing platforms.



(Trulia website listings page)

Architectural Model

Trulia is also most likely uses a modern web architecture based on MVC principles, with a React-based frontend communicating with RESTful backend services. At the data layer, it combines a relational database for structured listing data with big data processing tools for its Zestimate pricing algorithm. Zillow Group's acquisition of Trulia means the neighborhood review features are served from a separate but integrated service.

Core Features

- A large searchable database of homes for sale, rent, and recently sold
- "Zestimate" — an AI-generated estimate of a property's current market value
- Interactive map-based search with filters for schools, transport, and nearby amenities
- Neighborhood insights including crime data, school ratings, and walkability scores
- User reviews from residents about their neighborhoods
- A community Q&A section where users can ask questions about specific areas

What It Does Well

Trulia's most distinctive contribution is that it treats housing search as more than just a transaction. Through neighborhood ratings, resident reviews, and community Q&A features, it gives users a way to understand what it actually feels like to live somewhere — not just what the listing says. This is the community layer that makes the platform significantly more useful than a simple listings database.

What It Is Missing

Trulia was built for the American market and has no presence or relevance in China. Its community review model — which is genuinely valuable — does not exist in any Chinese housing platform. It also does not specifically address the needs of foreigners or international users trying to navigate a new country.

2.5 Summary and Comparison

Feature	Anjuke	Lianjia	Trulia	This Project
Housing listings	Yes	Yes	Yes	Yes
Verified listings	No	Yes	Partial	Partial
Tenant / resident reviews	No	No	Yes	Yes
Neighborhood ratings	No	No	Yes	Yes
English / multilingual support	No	No	English only	Yes
Foreigner-focused features	No	No	No	Yes
Community-sourced content	No	No	Partial	Yes
Relevant to China's market	Yes	Yes	No	Yes

Key Takeaway

All three platforms do certain things well — Anjuke has the listings volume, Lianjia has the verification and trust, and Trulia has the community review model. But none of them combine all three, and none of them serve the specific group this project is built for: foreigners and young people navigating China's housing market without local connections or Mandarin skills. This is the gap that this project directly targets.

Chapter 3 — Detailed Design of Personal Project Modules and Functions

3.1 Chosen Architecture Model

This project follows the Five-Tier Architecture pattern, which is the standard model taught throughout this course. The five tiers are :

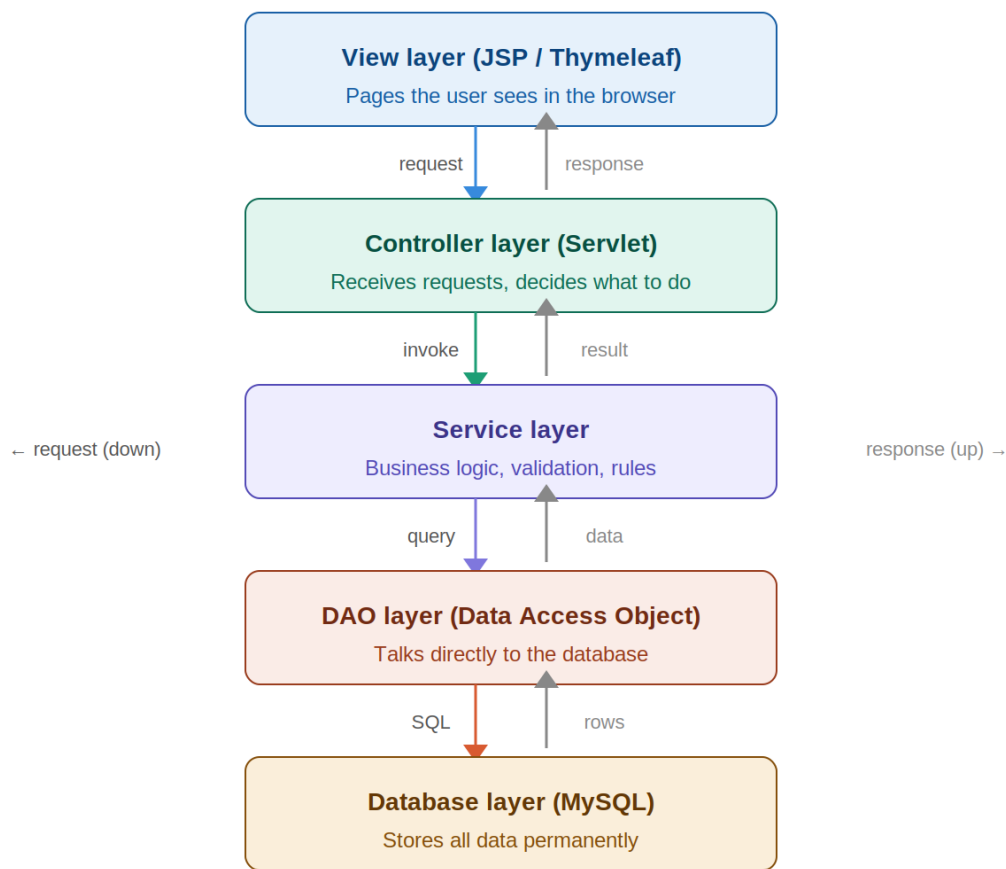
- View Layer (JSP / Thymeleaf) : the pages the user sees and interacts with in the browser
- Controller Layer (Servlet) : receives requests from the browser and decides what to do with them
- Service Layer : handles all the business logic, validation, and rules
- DAO Layer (Data Access Object) : handles all direct communication with the database
- Database Layer (MySQL) : stores all the data permanently

This model is also consistent with the MVC (Model-View-Controller) design pattern, which is one of the most widely used patterns in web development. In MVC terms, the JSP pages act as the View, the Servlet acts as the Controller, and the Service + DAO layers together form the Model. Entity classes serve as data carriers across all layers.

For one module (the Neighborhood Ratings feature), the project also uses Spring Boot with Thymeleaf as a modern alternative to JSP, demonstrating the full range of technologies covered in this course.

The execution flow for every user action follows this path:

JSP Page → Servlet → Service → DAO → MySQL → back up the same chain



(Five-Tier Architecture Flow)

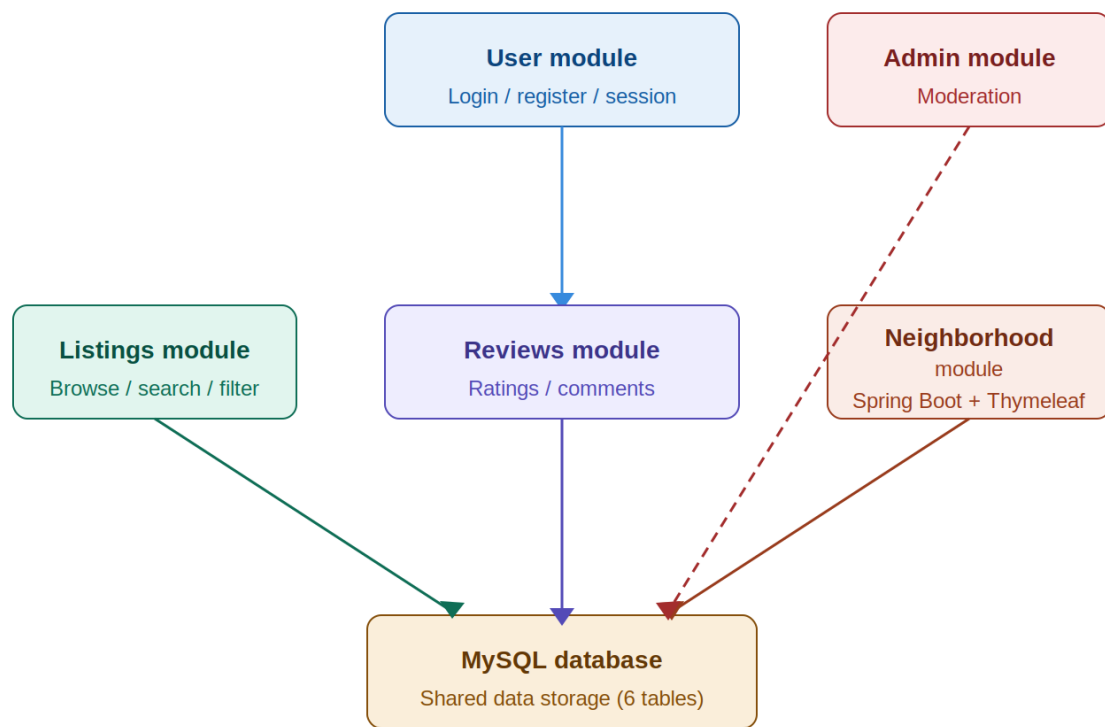
3.2 Overall System Architecture

The platform consists of five main modules:

- User Module
- Housing Listings Module
- Reviews and Ratings Module
- Neighborhood Module
- Admin Module

Each module follows the same five-tier structure and connects to a shared MySQL database.

The diagram below shows how the modules relate to each other and to the shared database.



All modules follow the five-tier architecture and share one MySQL database

(System Module Overview)

3.3 Module 1 — User Module

Overview

The User Module handles everything related to user identity — registering a new account, logging in, maintaining login state across pages, and logging out. This is the entry point for all other features since most actions on the platform require the user to be logged in.

Core Functions

- User Registration — new users create an account with username, email, password, nationality, and city
- User Login — registered users log in with their username and password
- Remember Me — users can stay logged in across browser sessions using a Cookie
- Login State Management — the user's identity is stored in a Session after login
- Logout — destroys the Session and clears the Remember Me Cookie
- Login Interception — a Filter redirects unauthenticated users to the login page

Implementation Logic

Registration uses a RegisterServlet that reads form parameters, calls UserService to validate input and check for duplicate usernames, then calls UserDAO to insert the new user into the database. Login uses a LoginServlet that verifies credentials through UserService and UserDAO, then stores the user object in the HttpSession. The Remember Me feature sets a Cookie with a 7-day expiry. A LoginFilter mapped to /* checks every incoming request for a valid Session. Logout is handled by LogoutServlet which calls session.invalidate() and sets the Cookie maxAge to 0.

Key Technologies

Servlet, HttpSession, Cookie, Filter, FilterChain, JSP, JDBC, DAO pattern, five-tier architecture

Database Table

```
user (id, username, email, password, nationality, city, role, created_at)
```

3.4 Module 2 — Housing Listings Module

Overview

The Housing Listings Module is the core browsing experience. It allows users to view available housing listings, search and filter by city, price range, and property type, and view the full details of any listing they are interested in.

Core Functions

- Browse Listings — paginated list of all available housing listings
- Search and Filter — filter by city, price range, and property type
- Listing Detail Page — full details including description, address, price, and average tenant rating
- Submit a Listing — registered users can submit a new housing listing
- My Listings — users can view and manage their own submitted listings

Implementation Logic

ListingServlet handles GET requests, reads filter parameters using getParameter(), and passes them to ListingService. ListingService validates the filter inputs and calls ListingDAO with the appropriate query parameters. ListingDAO builds a dynamic SQL query and returns a list of Listing entity objects. The JSP view uses JSTL c:forEach to loop through the list and EL

expressions like `${listing.price}` to display fields. Pagination is handled using SQL LIMIT and OFFSET.

Key Technologies

Servlet (doGet, doPost), getParameter(), JSP, EL expressions, JSTL c:forEach, JDBC PreparedStatement, DAO pattern, five-tier architecture

Database Table

```
listing (id, user_id, title, description, city, address, price,
property_type, created_at)
```

3.5 Module 3 — Reviews and Ratings Module

Overview

This is the most distinctive module of the platform. It allows tenants who have lived in a property to leave an honest written review and a star rating (1 to 5). This community-sourced layer is what separates this platform from existing Chinese housing apps.

Core Functions

- Submit a Review — logged-in users write a review and give a star rating for any listing
- View Reviews — all reviews for a listing are displayed on its detail page, newest first
- Average Rating Display — each listing shows an automatically calculated average star rating
- One Review Per User Per Listing — prevents spam and duplicate reviews
- Edit or Delete Own Review — users can update or remove their own reviews

Implementation Logic

ReviewServlet handles both GET (load reviews) and POST (submit review) requests. On POST, it reads listing_id, rating, and review_text from the request, calls ReviewService to validate, then ReviewDAO to insert. The average rating is calculated using SQL AVG(rating) grouped by listing_id. The one-review-per-user check is enforced in ReviewService before allowing a new submission. JSP displays reviews using c:forEach and shows star ratings using EL expressions.

Key Technologies

Servlet (doGet, doPost), JSP, EL, JSTL, JDBC, DAO, SQL AVG() aggregation, Session for user identity check

Database Table

```
review (id, listing_id, user_id, rating, review_text, created_at)
UNIQUE constraint on (listing_id, user_id) – one review per user per
listing
```

3.6 Module 4 — Neighborhood Module

Overview

The Neighborhood Module gives users a broader view of the areas where listings are located. Instead of just evaluating a single apartment, users can explore an entire neighborhood — reading community tips, checking liveability scores, and understanding what daily life in that area is actually like. This module is implemented using Spring Boot and Thymeleaf.

Core Functions

- Neighborhood Profile Page — dedicated page per neighborhood with name, city, description, and liveability score
- Liveability Ratings — users rate a neighborhood across four categories: safety, transport, food access, and foreigner-friendliness
- Community Tips — users post short text tips about a neighborhood
- Listings in This Neighborhood — each neighborhood page also shows all related housing listings

Implementation Logic

This module is built as a Spring Boot application with a NeighborhoodController using @GetMapping and @PostMapping annotations. The controller calls NeighborhoodService, which calls NeighborhoodRepository (DAO) to fetch data from MySQL. Thymeleaf templates use th:each to loop through listings and tips, and th:text to display values. Liveability scores are averaged across all user submissions per category.

Key Technologies

Spring Boot, @Controller, @GetMapping, @PostMapping, Model, Thymeleaf (th:each, th:text, th:action), JDBC, DAO pattern

Database Tables

```
neighborhood (id, name, city, description)
neighborhood_rating (id, neighborhood_id, user_id, safety, transport,
food_access, foreigner_friendly)
neighborhood_tip (id, neighborhood_id, user_id, tip_text, created_at)
```

3.7 Module 5 — Admin Module

Overview

The Admin Module gives a designated administrator account the ability to manage the platform's content — approving or removing listings, moderating reviews, and managing user accounts. This ensures the platform stays clean and trustworthy for all users.

Core Functions

- Admin Login — admin is identified by a role flag in the database and redirected to the admin dashboard after login
- Manage Listings — view all listings, approve pending ones, edit details, or delete inappropriate ones
- Moderate Reviews — view all reviews and delete any that violate community guidelines
- Manage Users — view all registered users and deactivate abusive accounts
- Dashboard Overview — shows key statistics: total users, listings, reviews, and pending listings

Implementation Logic

Admin identity is stored in the user table with a role column set to "admin". After login, LoginServlet checks the role and redirects to /admin/dashboard instead of the regular homepage. An AdminFilter is mapped to /admin/* and checks that the current session user has the admin role — any non-admin accessing /admin/* is immediately redirected. AdminDashboardServlet queries summary counts and passes them to the dashboard JSP via request.setAttribute(). Each admin management Servlet handles GET (list view) and POST (actions: delete, approve, deactivate) handlers.

Key Technologies

Servlet, AdminFilter on /admin/*, HttpSession role check, JSP, JSTL, EL, JDBC, DAO, five-tier architecture, request.setAttribute(), request forwarding

Database Addition

```
role VARCHAR(20) DEFAULT 'user' -- added to user table
Admin account seeded manually with role = 'admin'
```

3.8 Summary of All Database Tables

Table	Module	Purpose
user	User	Stores all registered user accounts
listing	Housing Listings	Stores all housing listings
review	Reviews & Ratings	Stores tenant reviews and star ratings
neighborhood	Neighborhood	Stores neighborhood profiles
neighborhood_rating	Neighborhood	Stores per-user liveability ratings
neighborhood_tip	Neighborhood	Stores community tips per neighborhood

3.9 Summary of Technology Mapping

J2EE Technology	Where It Is Used
HTTP / Request-Response	Every module — all user interactions go through HTTP
Servlet (doGet, doPost)	All modules — core request handler for every feature
Servlet lifecycle & @WebServlet	All Servlets use annotation-based configuration
getParameter / getParameterValues	Login, Register, Search filters, Review submission
Request forwarding	Admin dashboard passing data to JSP
Redirect (sendRedirect)	After login, logout, and form submission success
Session + JSESSIONID	User login state maintained across all pages
Cookie	Remember Me feature in User Module
Filter + FilterChain	LoginFilter (all pages) and AdminFilter (/admin/*)
web.xml filter order	StaticFilter before LoginFilter before AdminFilter
JSP	All view pages
EL Expressions (\${...})	Displaying data in all JSP pages
JSTL (c:forEach, c:if)	Looping listings, reviews, conditional display
JDBC + PreparedStatement	All DAO classes for database operations
JdbcUtil / DAO pattern	Shared utility class across all modules
Five-Tier Architecture	Entire project structure
Spring Boot @Controller	Neighborhood Module
Thymeleaf (th:each, th:text)	Neighborhood Module view layer
MySQL	All data storage